EverConnect: Internet Service Churn Analysis

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EXECUTIVE SUMMARY

Project Description

This analysis aims to identify key factors driving customer churn at EverConnect and offer actionable insights to improve retention.

Objective

- Identify churn drivers.
- Analyze customer behavior patterns.
- Provide recommendations to reduce churn and improve retention.

Tools

- Python
- Tableau
- Google Spreadsheets (initial data cleaning)

Metrics

- Churn Rate
- ARPU
- CLV

Analysis Result

- Found key drivers of churn
- Generated a good predictive model
- Provided customer segmentation

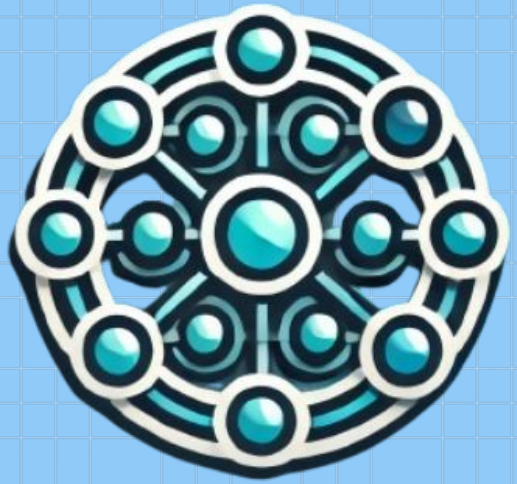
Recommendation

- Target TV Basic TV Subscribers
- Engage with "Heavy Data Users"
- Monitor "High Churn, No TV Users"

EverConnect Background

A telecommunications company focused on providing high-quality internet and communication services to both residential and business customers.

EverConnect is aiming to reduce customer churn by understanding the key factors using data-driven insights to enhance its strategies.



OVERVIEW : INTERNET SERVICE CHURN DATASET

Source
The dataset used in this analysis was sourced from **Kaggle**. Understanding churn and retention is vital for implementing strategies to enhance customer loyalty and address business challenges.
Contents
This dataset includes key features that help **identify factors contributing to churn and retention**. It was provided in CSV format and required cleaning for analysis.
Original Creator: **Mehmet Sabri Kunt**

Methodology

Step 1

Understanding business problem and defining the objective

Step 2

Researching in advance to further understand the content of the data

Step 3

Utilizing Spreadsheets and Python for data cleaning

Step 4

Data analysis with Python and Tableau

Step 5

Generating insights and providing recommendation

Problem Statement

How to reduce churn rate by 15% within 12 months by identifying and addressing the key factors driving churn?

Objectives

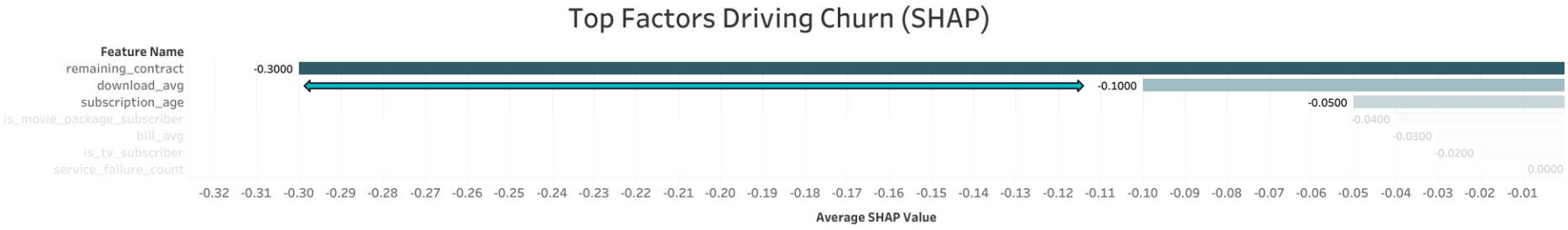
1. **Analyze Churn Drivers:** Identify key factors influencing customer churn using data-driven analysis and machine learning insights.
2. **Segment Customers:** Differentiate between churned and retained customers to better understand their behavior and needs.
3. **Prioritize Retention Strategies:** Provide actionable recommendations to reduce churn and enhance customer retention efforts.
4. **Optimize CLV:** Leverage insights to maximize customer lifetime value and improve overall profitability.

Hypothesis & Metrics

Hypothesis: Higher churn rates at EverConnect are primarily driven by factors such as subscription type, service issues, and higher bill averages, and addressing these factors will significantly reduce churn.

Metrics: Churn Rate, Customer Lifetime Value, Average Revenue Per User

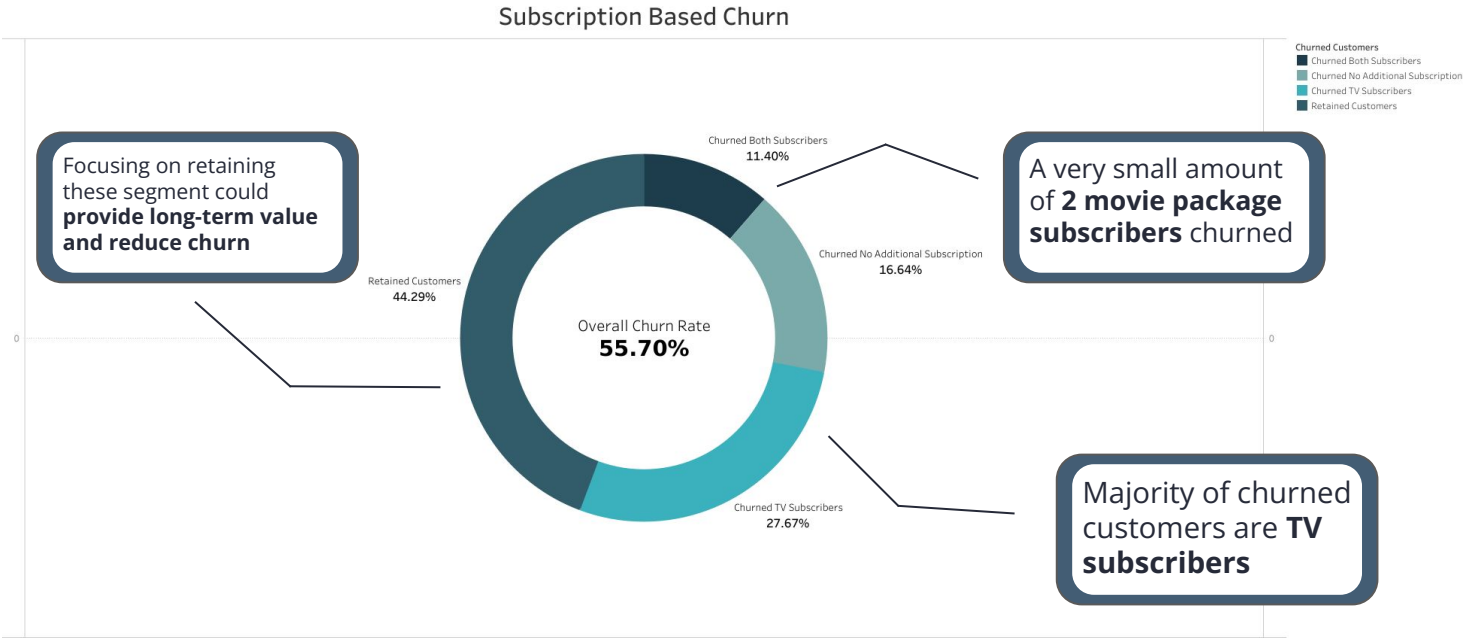
Key Factors Driving Churn



This chart highlights the most significant factors contributing to customer churn, ranked by their impact.

- **Remaining Contract**: Customers with fewer years left on their contract are more likely to churn.
- **Bill Average**: Higher billing amounts in the last three months correlate with churn.
- **Subscription Age**: Shorter subscription durations are linked to higher churn rates.

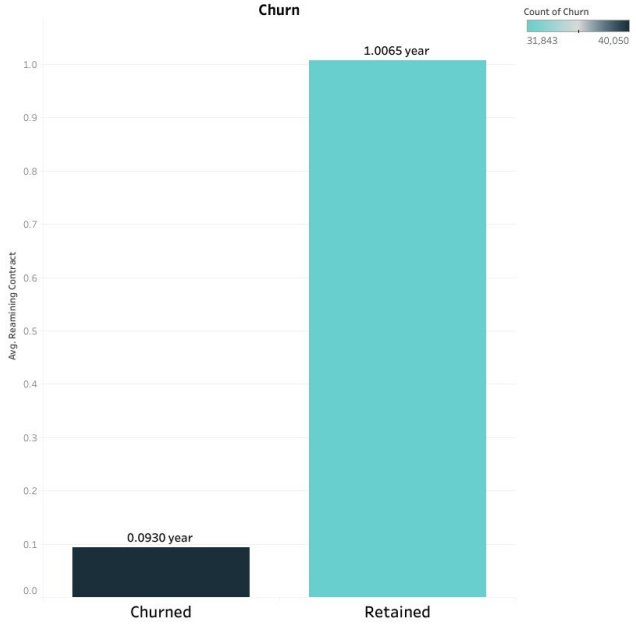
Churn Breakdown by Subscriptions



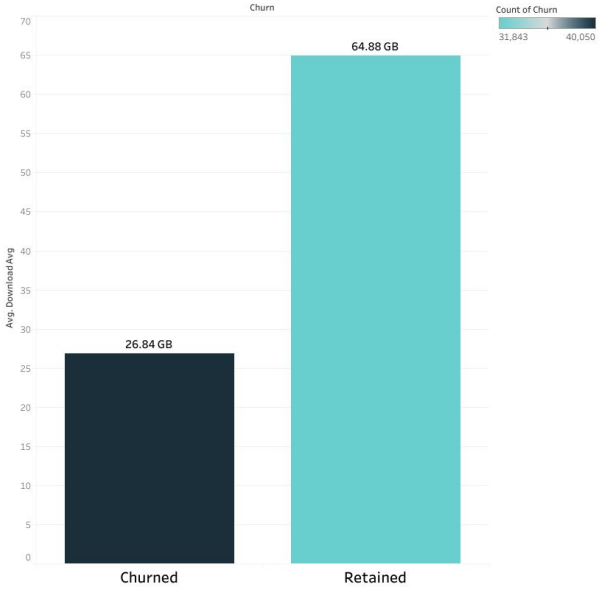
- **For TV subscribers**: Target this group with personalized retention strategies.
- **For movie package subscriber**: The low churn rate for movie package subscribers suggests lower risk in this segment, but further analysis could help confirm why this is the case.
- **For retained customer**: These customers have a higher lifetime value.

Impact of Contract and Download Usage on Churn

Remaining Contract Impact On Churn



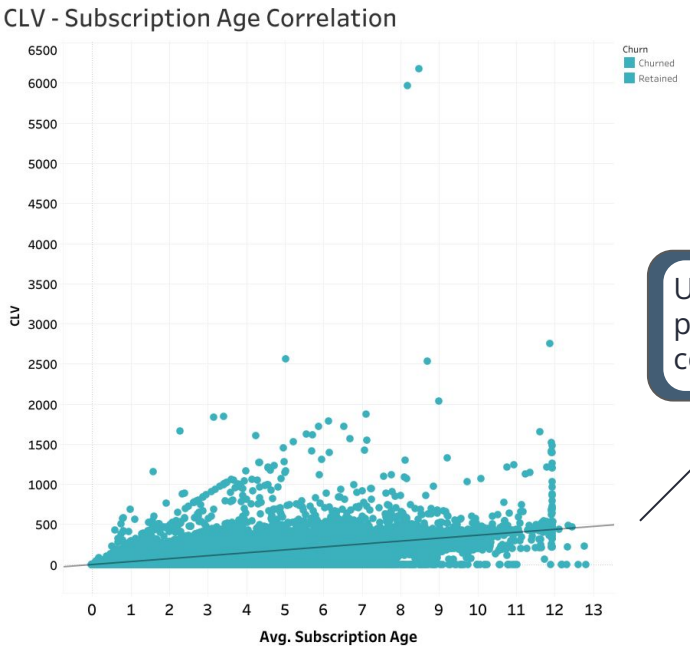
Average Download Usage Impact On Churn



Customers with **fewer year contract tend to churn.**

Shows a clear uptrend **as high usage increase, customers tend to stay and keep using the service.**

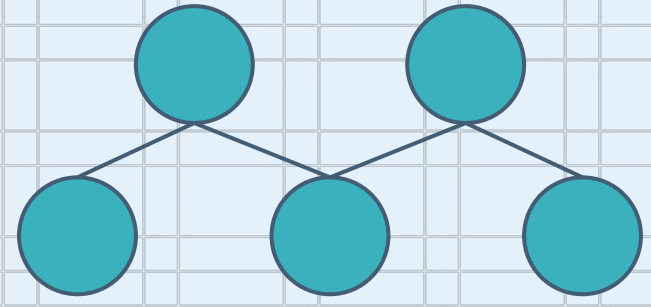
Subscription Age Correlation with CLV



Uptrend, shows positive correlation.

- **Positive Correlation**: Longer subscription age, higher CLV.
- **Retention Opportunity**: Can significantly boost revenue.

Churn Prediction Random Forest



93.41%
Accuracy

1. Remaining_contract
2. Bill_avg
3. Service_failure_count
4. Subscription_age
5. Download_avg
6. Is_tv_subscriber
7. is_movie_package_subscriber

96.2%
ROC-AUC Score

Objective: To predict customer churn based on the dataset.

Customer Segmentation K-means

Cluster	Name	Key Characteristics	Churn Rate
0	Basic TV Subscribers	Loyal TV-only customers, low download usage, moderate bill average	57.45%
1	Engaged Movie Package Users	High engagement with movie package, moderate download usage, low churn	33.99%
2	Heavy Data Users with Service Issues	High download usage, service issues, high bill average, moderate churn	55.67%
3	High Churn, No TV users	No TV subscription, low usage, high churn rate	90.59%

Customer Segmentation K-means

- **Cluster 0**: Focus on offering additional services to enhance their experience and reduce churn.
- **Cluster 1**: Highly engaged customers, so maintaining and expanding the movie package offerings to retain.
- **Cluster 2**: Addressing service quality and improving network reliability for this segment could significantly reduce churn.
- **Cluster 3**: Personalize offer or re-engage campaigns to either retain or identify the reasons behind their high churn.

Overall Insights

1. **Churn is Highly Influenced by Subscription Type:**
- TV subscribers have a significantly higher churn rate compared to movie package subscribers.
- Customers who do not have a TV subscription (Cluster 3) are at an extremely high churn risk.

2. **Bill Average and Usage Correlate with Churn Risk:**
- Customers with lower bill averages and lower average download usage tend to have higher churn rates.
- Higher bill averages and greater download usage are associated with higher CLV and lower churn risk.

3. **Subscription Age Plays a Key Role:**
- Longer subscription age correlates with better customer retention and higher CLV.
- Newer customers (shorter subscription age) are more likely to churn, especially if they are in clusters with low engagement.

Overall Insights

4. Customer Segmentation Reveals Distinct Churn Profiles:

- **Cluster 0 (Basic TV Subscribers):** Moderate churn risk. These customers have been with the company for a while but show moderate usage and bill averages.
- **Cluster 1 (Engaged Movie Package Users):** Low churn. These customers are highly engaged, with long subscriptions and high download usage.
- **Cluster 2 (Heavy Data Users with Service Issues):** Moderate churn risk, but high download usage and service issues could drive dissatisfaction.
- **Cluster 3 (High Churn, No TV Users):** Very high churn. These customers lack TV subscriptions and have low usage. They are the most disengaged group.

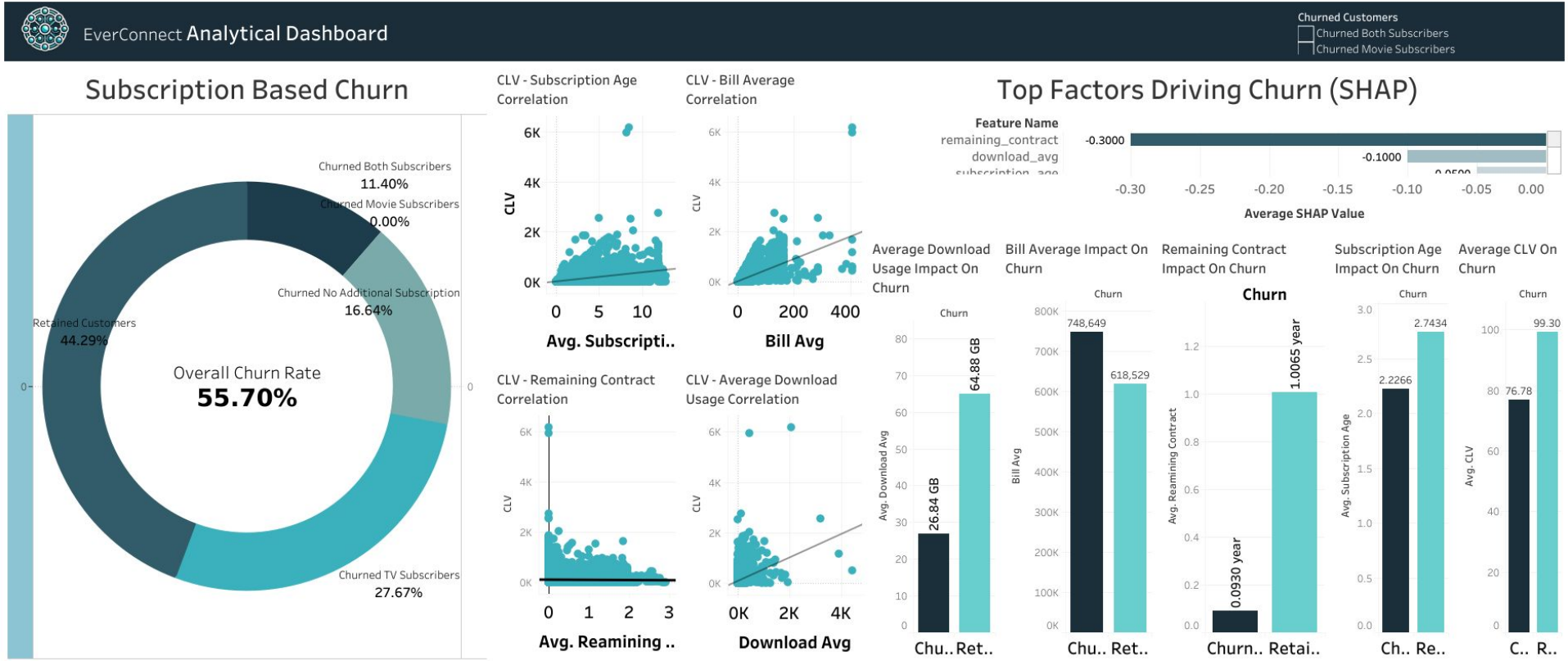
5. Predictive Modeling Insights:

- The Random Forest model achieved high predictive accuracy (93.4%) and ROC-AUC score (96.2%), suggesting that churn prediction is highly reliable.
- SHAP analysis revealed that subscription type, bill average, and download usage are the top factors driving churn, offering key areas to focus on for retention strategies.

Recommendation

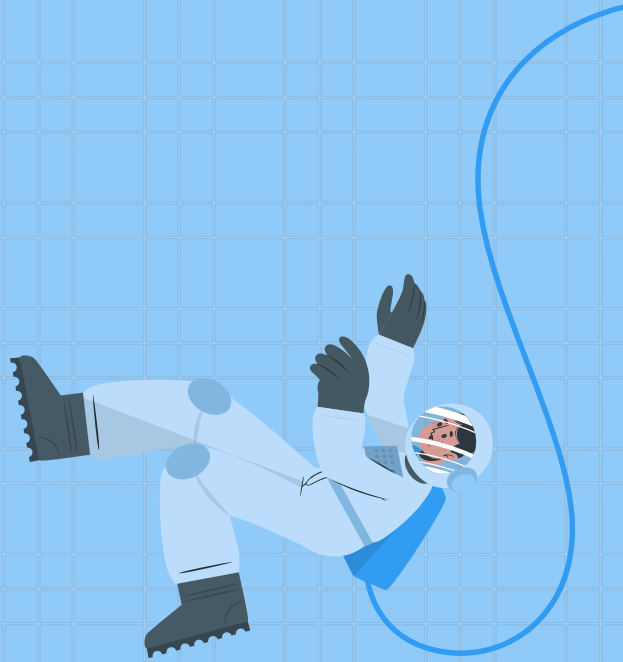
1. **Focus on Retaining TV Subscribers:**
   - Develop loyalty programs or bundle services for Basic TV Subscribers (Cluster 0).
   - Offer TV subscriptions to No TV Users (Cluster 3) to reduce their high churn risk.
2. **Engage Movie Package Users:**
   - Provide personalized offers and exclusive content for Engaged Movie Package Users (Cluster 1) to maintain their low churn rate.
3. **Improve Service Quality for Heavy Data Users:**
   - Address service issues for Heavy Data Users (Cluster 2) to reduce churn and improve customer satisfaction.
4. **Tailor Plans Based on Usage:**
   - Offer customized plans for low-usage customers to increase engagement and value.
5. **Boost Retention for New Customers:**
   - Implement onboarding programs and personalized promotions to engage newer customers and reduce churn.
6. **Leverage Predictive Models for Proactive Retention:**
   - Use churn prediction models to identify at-risk customers and apply targeted retention strategies.

Overall Tableau Public Dashboard

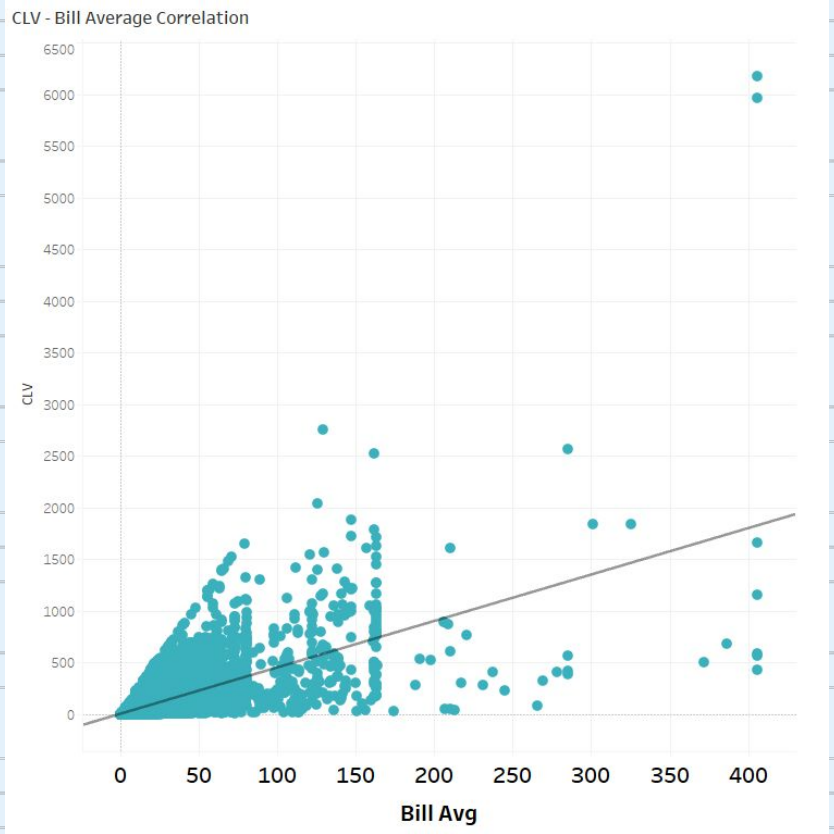


[Link to Dashboard here]

Appendix



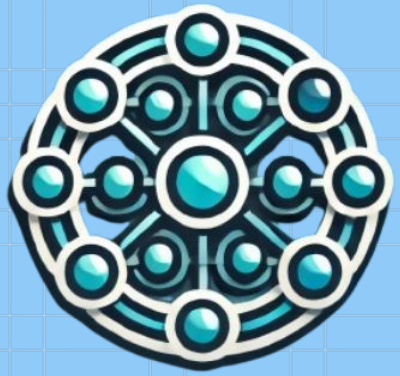
Bill Avg - CLV Correlation



High CLV definitely drives revenue. Positive correlation.

Additional Data

Dataset link: https://www.kaggle.com/datasets/mehmetsabrikunt/internet-service-churn

Google Colab link: https://colab.research.google.com/drive/1AH-0UCggpVrqoJfbJn-VTtg-lgVFCNEU?usp=sharing

Thanks!